



RAJALAKSHMI ENGINEERING COLLEGE

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Experiment 3: Setting Up Hybrid Cloud Connectivity Using Cisco Packet Tracer (Simulation)

Aim

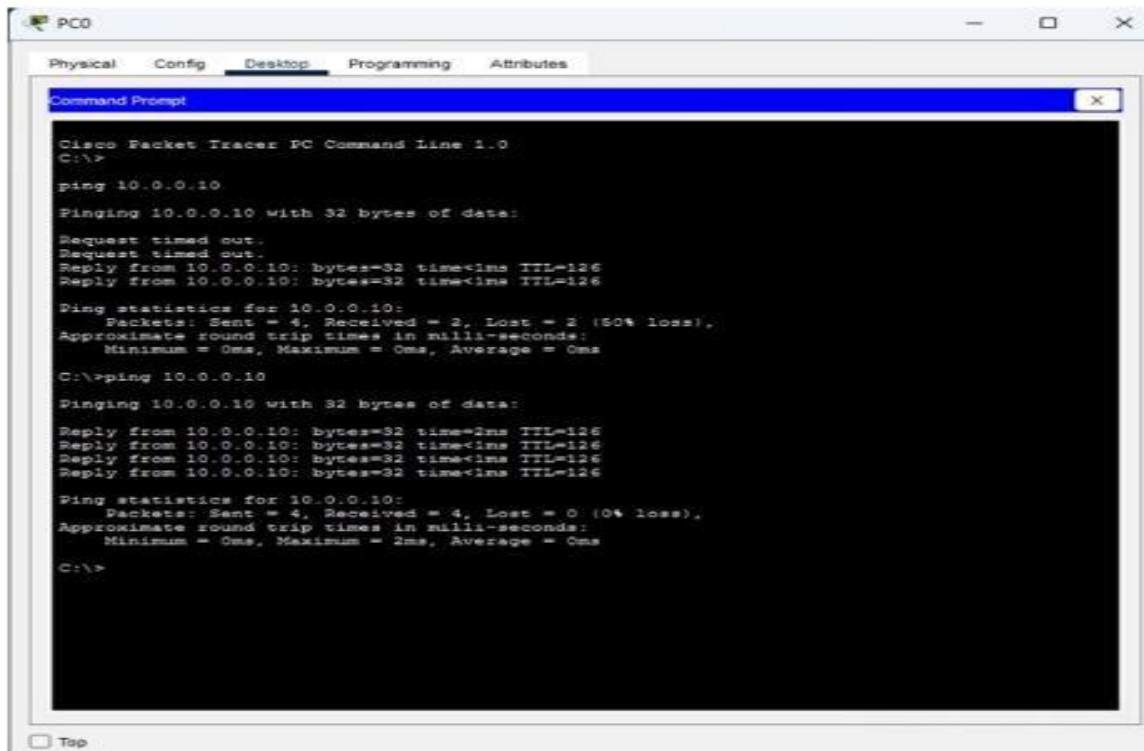
To simulate a secure hybrid cloud connection between an on-premises data center and a cloud network using routers, switches, subnets, and routing in Cisco Packet Tracer.

INTRODUCTION

A hybrid cloud environment integrates an organization's on-premises network with a cloud-based network, allowing secure and efficient data exchange between both environments. In this experiment, Cisco Packet Tracer is used to design and simulate a hybrid cloud network consisting of routers, switches, a PC, and a server. Static routing is configured to establish communication between the local network and the cloud network, ensuring successful end-to-end connectivity.

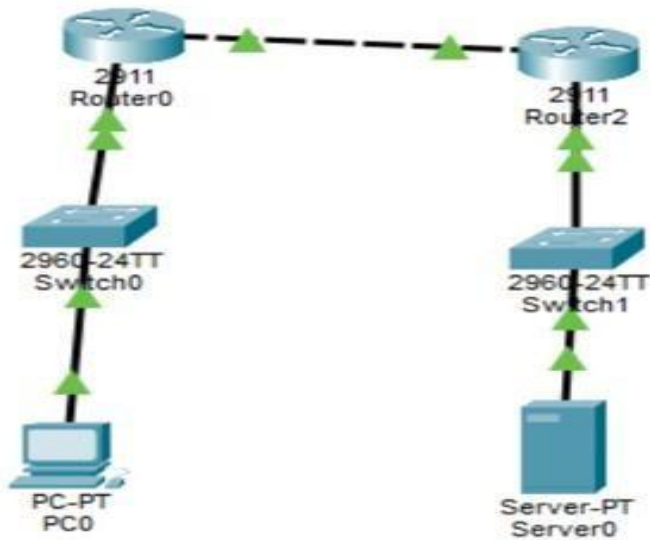
Procedure / Steps

1. Open Cisco Packet Tracer and create a network topology by placing two routers, two switches, one PC, and one server in the workspace.
2. Connect all devices using the appropriate cables according to the network design.
3. Configure IP addresses for the router interfaces and enable them using the no shutdown command.
4. Assign IP addresses and default gateway settings to the PC and the server.
5. Configure static routing on both routers to establish communication between the two networks.
6. Save the router configurations and verify the routing tables using the show ip route command.
7. Test network connectivity by pinging the server from the PC.
8. Confirm successful end-to-end communication between the on-premises network and the cloud network.



The screenshot shows a Cisco Packet Tracer PC Command Prompt window. The window title is "PC0" and it has tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, showing a "Command Prompt" window. The command prompt displays the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>
ping 10.0.0.10
Pinging 10.0.0.10 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 10.0.0.10: bytes=32 time<1ms TTL=126
Reply from 10.0.0.10: bytes=32 time<1ms TTL=126
Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 10.0.0.10
Pinging 10.0.0.10 with 32 bytes of data:
Reply from 10.0.0.10: bytes=32 time=2ms TTL=126
Reply from 10.0.0.10: bytes=32 time<1ms TTL=126
Reply from 10.0.0.10: bytes=32 time<1ms TTL=126
Reply from 10.0.0.10: bytes=32 time<1ms TTL=126
Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms
C:\>
```



Result

A hybrid cloud network was successfully simulated using Cisco Packet Tracer. The on-premises network and cloud network were connected through routers using static routing. Successful ping responses between the PC and the server confirmed end-to-end network connectivity.